
Evidence-Blindness in Multimodal Medical Claim Verification: A Counterfactual Diagnostic and Training-Time Mitigation

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Abstract

Multimodal claim verifiers are deployed as a safety layer downstream of chest-X-ray report generators, and recent systems report above 90% AUROC on hallucination-detection benchmarks. We show those numbers can be reached by a verifier that never looks at the image — a failure mode we call evidence-blindness. To detect it, we propose three counterfactual perturbations of a verifier’s input and measure how much accuracy drops: zeroing the image (IMG), shuffling the retrieved evidence across claims (ESG), and flipping the image left-to-right on laterality-sensitive claims (IPG). The diagnostic immediately catches a trap: the verifier in our study with the highest validation accuracy turns out to be fully evidence-blind on the IMG axis. A consistency loss combined with an adversarial hypothesis-only filter closes the gap to roughly 70pp at matched accuracy. The fix is brittle, and the same diagnostic catches it: under leave-one-site-out training, the gap regresses to baseline on the held-out site. We further wrap inverted conformal selection around each verifier and find that every evidence-blind configuration produces an empty trusted set; only the non-blind verifier yields a populated one. A conformal guarantee on an evidence-blind verifier has nothing to certify. We argue that counterfactual sensitivity testing should precede both aggregate-accuracy reporting and conformal trust wrapping for any multimodal medical verifier.

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1. Introduction

Radiology report generation has moved from a demonstration task to a near-deployment tool. Systems such as MAIRA-2 (Bannur et al., 2024), CheXagent-2 (Chen et al., 2024), and MedGemma (Google HAI-DEF, 2025) produce structured findings reports from chest radiographs that are beginning to reach radiologists’ review queues as first drafts, with reported hallucination rates of 8–15% (Zhang et al., 2025; Hardy et al., 2025). The community’s response is to insert a claim-level verifier downstream of the generator. Recent verifiers report AUROC above 90% on aggregate hallucination-detection benchmarks, and the conversation has shifted from whether claim verification works at all to which conformal guarantee should wrap it (Mohri & Hashimoto, 2024; Angelopoulos & Bates, 2021; Gui et al., 2024; Li et al., 2025).

Aggregate accuracy on a held-out test set cannot distinguish a verifier that uses the image from one that exploits textual shortcuts in the claim and the retrieved evidence. Gururangan et al. (2018) and Poliak et al. (2018) established the analogous finding for natural language inference: hypothesis-only classifiers reached competitive accuracy on SNLI and MultiNLI without observing the premise, and the diagnostic reshaped NLI evaluation practice. The radiology-specific instance of this question is most directly addressed by Restrepo et al. (2025), who introduce Selective Modality Shifting (SMS) on MIMIC-CXR and FairVLMed and report a marked dependency on text input across four generalist VLMs and two medical-specialist VLMs. Restrepo et al. explicitly leave training-time mitigation to future work; the conformal-selection literature similarly wraps an existing predictor without modifying it.

We close two gaps. (i) We define evidence-blindness via three counterfactual metrics — IMG, ESG, IPG — that apply to any binary multimodal verifier including closed-weight APIs, and report a laterality-flip variant (IPG) that does not appear in the SMS family. (ii) We train a cross-modal verifier on a public, non-credentialed chest-radiograph benchmark and show that

its highest- validation-accuracy configuration is fully evidence-blind, while a consistency loss combined with an adversarial hypothesis-only filter lifts IMG from 2.03 pp to 69.24 pp at the same test accuracy. We report a third empirical observation: an inverted conformal Benjamini– Hochberg selection procedure collapses to an empty trusted set on every evidence-blind configuration and produces a populated trusted set only on the non-blind one — placing the counterfactual diagnostic strictly before conformal wrapping in any deployment pipeline.

2. Methods and Benchmark

Counterfactual diagnostics. Let f be a binary multimodal verifier mapping $(x_{\text{img}}, x_{\text{claim}}, x_{\text{evid}}) \rightarrow \{\text{sup}, \text{con}\}$. For a labelled evaluation set $\mathcal{D}$:

$$\text{IMG}(f) = \text{acc}(f, \mathcal{D}) - \text{acc}(f, \mathcal{D}^{\text{img}=0}), \quad (1)$$

$$\text{ESG}(f) = \text{acc}(f, \mathcal{D}) - \mathbb{E}_{\pi}[\text{acc}(f, \mathcal{D}^{\text{evid}=\pi})], \quad (2)$$

$$\text{IPG}(f) = \text{acc}(f, \mathcal{D}_{\text{lat}}) - \text{acc}(f, \mathcal{D}_{\text{lat}}^{\text{flip}}), \quad (3)$$

in percentage points, where $\mathcal{D}^{\text{img}=0}$ replaces each image with a zero tensor, $\mathcal{D}^{\text{evid}=\pi}$ replaces each evidence text with $x_{\text{evid}}^{(\pi(i))}$ under a within-batch derangement π (averaged over three seeds), and $\mathcal{D}_{\text{lat}}^{\text{flip}}$ horizontally flips images on the subset of claims matching the regular expression `\b(left|right|bilateral)\b`. We treat f as evidence-blind if $\text{IMG}(f) < 5$ pp or $\text{ESG}(f) < 5$ pp; threshold sensitivity over $\{3, 5, 7, 10\}$ pp is reported in Appendix ???. The diagnostic requires only input substitution and applies to closed-weight verifiers exposed via API.

Cross-modal verifier. Images at 224×224 pass through BiomedCLIP ViT-B/16 (Zhang et al., 2024); the top four of twelve transformer blocks are trainable. Claim and evidence tokens are jointly encoded by RoBERTa-large (Liu et al., 2019) via the standard text- pair API; the top eight of twenty-four layers are trainable. Both modalities project to a shared dimension $d = 768$. A learnable [VERDICT] token is prepended at position zero of the fused sequence, which then passes through four bidirectional cross-modal transformer layers with twelve heads each. Three heads attach to the verdict-token output: a binary verdict head (2-way softmax), a support-score head (sigmoid scalar, used downstream for conformal calibration), and a 14×14 per-patch grounding head (sigmoid map). The model has ≈ 480 M total parameters of which 122 M are trainable. Optimization details and the full loss equation are in Appendix ??.

Composite loss and hypothesis-only filter. The training loss is a five-term composite: classification cross-entropy ($\mathcal{L}_{\text{cls}}$), bounding-box-supervised patch grounding ($\mathcal{L}_{\text{ground}}, \lambda = 0.5$), an image-masked consistency margin $\mathcal{L}_{\text{consist}} = \text{ReLU}(0.2 - (p_{\text{full}}(y) - p_{\text{img}=0}(y)))$ ($\lambda = 0.3$), a contrastive evidence margin restricted to sup anchors ($\lambda = 0.3$), and an MC-dropout uncertainty regulariser ($\lambda = 0.1$). The adversarial hypothesis-only (HO) filter is a separate RoBERTa-large cross-encoder trained for one epoch on $(\text{claim}, \text{evidence}) \rightarrow y$ pairs with no image input. At training time, rows on which the HO filter’s gold-class probability exceeds 0.7 have their $\mathcal{L}_{\text{cls}}$ term downweighted by $0.2\times$; the other four terms retain full weight. Loss-drop ablations (Appendix ??) show $\mathcal{L}_{\text{consist}}$ and the HO filter are the two load-bearing terms: removing either collapses IMG to ≤ 6 pp.

Benchmark. GroundBench-v6 assembles three public sources: OpenI (Demner-Fushman et al., 2016) ($\sim 4,000$ US frontal radiographs), ChestX-Det10 (Liu et al., 2020) ($\sim 3,500$ with pixel annotations), and PadChest-GR (Castro et al., 2024) (4,555 bilingual studies with per-sentence radiologist boxes). Patient-stratified splits at 70/10/10/10 yield 28,361 train, 4,046 validation, 4,046 calibration, and 4,132 test claims. For real-hallucination evaluation, 1,500 findings reports from MAIRA-2 (Bannur et al., 2024), CheXagent-2-3B (Chen et al., 2024), and MedGemma-4B-IT (Google HAI-DEF, 2025) are decomposed into 2,707 atomic claims and silver-labelled by a three-LLM judge ensemble (Ostmeier et al., 2024). Detail in Appendix ??.

3. Results

The classification baseline is evidence-blind. Table 1 reports the diagnostic on the 2,906-claim in-distribution test split. The cross-entropy baseline v5.0 reaches 90.9% test accuracy yet IMG is only 2.03 pp; v5.1 adds patch grounding, achieves the highest validation accuracy (91.1%), and remains evidence-blind at $\text{IMG} = 2.17$ pp. A practitioner selecting on aggregate accuracy alone would ship the most blind of the configurations we trained.

Consistency + HO filter is the load-bearing intervention. v5.2 adds both. IMG jumps from 2.17 to 62.90 pp; the zero-image accuracy collapses to 29.4% (below random) because the consistency loss specifically rewards divergence under image masking, while aggregate accuracy rises to 92.3%. Adding contrastive evidence extends IMG to 69.24 pp at 92.5% test accuracy. A confirmatory v6.0 retrain reaches 70.21 pp.

Table 1. Counterfactual diagnostic. Top: in-distribution test split. Bottom: cross-site leave-one-out (the v5.3 loss composition retrained for two epochs on the non-held-out sites and evaluated on the held-out site). A configuration is evidence-blind if $\text{IMG} < 5$ pp or $\text{ESG} < 5$ pp.

Config	Test acc	IMG	ESG	Blind?
v5.0-base	0.909	2.03	19.71	yes
v5.1-ground	0.921	2.17	21.13	yes
v5.2-real	0.923	62.90	18.74	no
v5.3-contrast	0.925	69.24	18.56	no
v6.0-retrain	0.911	70.21	20.01	no
Cross-site leave-one-out (held-out site):				
held out: OpenI	0.853	3.21	-0.02	yes
held out: ChestX-Det10	0.660	28.41	2.69	yes

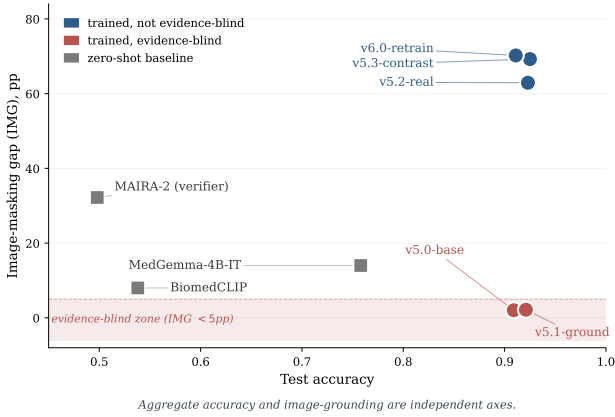


Figure 1. Aggregate accuracy and image-grounding are independent axes. Circles: trained configurations. Squares: zero-shot detectors on a 500-claim real-RRG subset. Shaded band: evidence-blind zone ($\text{IMG} < 5$ pp). The highest-validation-accuracy configurations (v5.0, v5.1) sit in the blind zone at 90–92% accuracy; zero-shot baselines BiomedCLIP and MedGemma-4B-IT are blind on the ESG axis at intermediate accuracy.

Field audit on real-RRG hallucinations. We extend the diagnostic to 500 silver-labelled real-RRG claims with four zero-shot detectors (Fig. 1; full numbers in Appendix ??). BiomedCLIP zero-shot reaches 0.538 accuracy with $\text{IMG} = 8.0$ pp and $\text{ESG} = 0.0$ pp (blind on ESG). MedGemma-4B-IT used as a verifier reaches 0.758 with $\text{IMG} = 14.0$ pp and $\text{ESG} = 2.0$ pp (blind on ESG). MAIRA-2 used zero-shot clears both thresholds ($\text{IMG} = 32.2$ pp, $\text{ESG} = 6.6$ pp) but at near-chance accuracy (0.498). The combination matters: a verifier can fail one axis without failing the other, and aggregate accuracy alone does not predict the joint outcome.

Cross-site regression and deployment alarm. Two leave-one-site-out configurations trained under the v5.3 loss composition show the mitigation does not transfer (Table 1, bottom). Held-out OpenI regresses

to $\text{IMG} = 3.21$ pp (within 1.2 pp of the v5.0 baseline); held-out ChestX-Det10 keeps most of the IMG mitigation but drops to $\text{ESG} = 2.69$ pp, below threshold. The same three metrics that catch training-time blindness catch post-deployment blindness. This is the second role of the same diagnostic.

Conformal selection collapses on blind verifiers. We wrap an inverted conformal Benjamini–Hochberg procedure around each verifier’s support score, calibrated on a held-out 4,046-claim calibration split at target $\alpha = 0.10$. Only v5.3-contrast produces a populated trusted set: $n_{\text{green}} = 985$ claims with achieved FDR = 0.91% and power 39.6%. The configurations v5.0, v5.1, and v5.2 collapse to $n_{\text{green}} = 0$; v5.0 and v5.1 collapse because they are evidence-blind, and v5.2 collapses because the joint distribution of (support score, true label) is not separable enough below the target level (Appendix ??). A conformal guarantee on a blind verifier has nothing to certify.

4. Discussion

The cross-entropy baseline reports a competitive 90.9% test accuracy while being image-blind at $\text{IMG} = 2.03$ pp. The consistency loss and HO filter act on disjoint failure modes: the consistency loss penalises verdict-invariance under image masking, and the HO filter downweights training rows the text-only model already solves. Adding patch grounding alone (v5.1) does not move IMG, mirroring NLI partial-input findings (Gururangan et al., 2018; Poliak et al., 2018) that auxiliary supervision on input attribution does not by itself prevent shortcut exploitation.

The mitigation indexes on training-distribution shortcuts. Under $\text{site}_{\text{train}} \neq \text{site}_{\text{test}}$, the shortcut structure changes and the mitigation has nothing to suppress — but the same diagnostic catches the post-shift blindness. This makes the counterfactual diagnostic the appropriate alarm signal in both training-time selection and deployment-time monitoring; the appropriate fix for the cross-site regime is distribution-shift-aware methodology (e.g., weighted conformal inference under covariate shift (Tibshirani et al., 2019)) layered on top of the same diagnostic.

The conformal-selection collapse on blind verifiers sharpens the Mohri–Hashimoto conformal-factuality guarantee (Mohri & Hashimoto, 2024): marginal coverage is preserved on whatever signal the underlying verifier emits, but if that signal is image-blind the resulting trusted set has no clinical content. Wrapping a frozen verifier in conformal selection is necessary but not sufficient for trustworthy deployment.

5. Limitations and Conclusion

Three limitations bound our claims. (L1) Scale. The benchmark spans approximately 7,500 images and 25,176 training claims — enough to expose evidence-blindness as a measurable failure mode but not to estimate its prevalence at population scale. (L2) Single-seed training. All training runs use a single random seed; multi-seed replication is deferred to an extended version. (L3) Spatial residual. Laterality-turning claims are insensitive to horizontal flips (IPG within $[-0.88, +0.88]$ pp across all configurations); the consistency loss targets image presence rather than spatial arrangement, and closing this residual would require an image-encoder-level intervention.

We introduced a counterfactual diagnostic for multimodal medical claim verification (IMG, ESG, IPG), a training-time recipe that closes the in-distribution IMG from 2.03 pp to 69.24 pp at +1.6 pp accuracy, and the empirical observation that inverted conformal selection collapses to an empty trusted set on every evidence-blind verifier we tested. Counterfactual evidence-sensitivity testing should precede aggregate-accuracy reporting and conformal trust-selection for any multimodal medical verifier. Benchmark maintainers and submission checklists in this area should report partial-input baselines alongside aggregate accuracy — as the NLI community did after Gururangan et al. (2018) and Poliak et al. (2018).

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